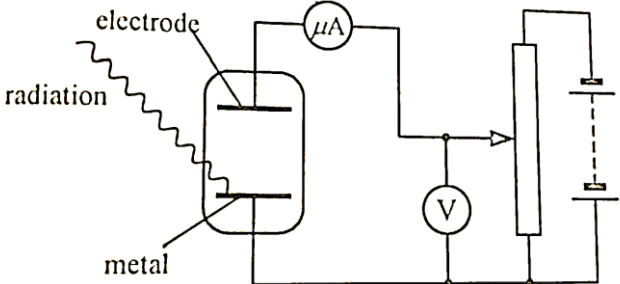


6(a)	 Photoelectric effect refers to the phenomenon that when some <u>metal surfaces are illuminated</u> by light, <u>electrons are emitted</u>. When the metal surface is illuminated with light even at a very <u>low intensity</u>, the flow of current in the ammeter indicates that <u>photoelectrons are emitted almost instantaneously</u>, as long as light is <u>above a threshold frequency</u>. This led to the idea that light exists as <u>particle-like photons</u>, each having a quantum of energy which can be absorbed by a an electron in the metal surface, releasing the electron to move towards the electrode.	B1 B1 B1
6(b)(i)	The energy of a photoelectron can be increase by using radiation with <u>higher frequency</u>. This is because each electron only absorbs <u>energy from one photon</u>.	B1
6(b)(ii)	The rate of production of photoelectrons can be increased by <u>increasing the intensity</u> of the radiation. This is because it is proportional to the <u>rate of photons arriving at the metal</u> which is in turn proportional to the intensity of the radiation.	B1 B1
6(c)	The <u>diffraction of light</u> demonstrates the wave nature of electromagnetic radiation, which is inconsistent with the evidence shown by the photoelectric effect. Diffraction is the <u>bending and spreading of waves when it passes through an aperture</u>, giving raise to a wider image on the screen than the aperture. <u>Only waves have this property</u>. Particles would only travel in a straight line, thus resulting in an image the same size as the aperture. OR ANY OTHER EXPERIMENTS THAT PROVE WAVE NATURE OF LIGHT.	B1 B1 B1